

An integrated metric for rapid and equitable emergency rescue during urban flash flooding events

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ABSTRACT

During flash flooding, quick and effective rescue operations are crucial to minimizing harm to vulnerable communities. While much research focused on emergency response and evacuation, few studies address how overhead powerline obstructions impact rescue operations. Additionally, existing research on vulnerable communities often emphasizes long-term flood mitigation and recovery but less so on immediate responses. To ensure rapid and equitable flood rescue operations, this study derives an integrated metric to quantify rescue demands that incorporate rescue efficiency, community flood severity, and social vulnerability. In detail, rescue efficiency is calculated by analyzing a network that captures the geospatial interdependencies between the residential buildings' road networks and overhead power lines; community flood severity is quantified as the percentage of building damage resulting from flood impacts; and social vulnerability is an integrated indication of key household composition factors (e.g., elders, single parents, and minorities). Based on this metric, a systematic step is designed to suggest the sequence of rescue operations and the strategies for distributing rescue lifeboats at emergency facilities. The applicability and feasibility of the proposed approach were demonstrated using lifeboat rescue operations in Manville, New Jersey, during Hurricane Ida. This study calculates dynamic changes in rescue loads of all emergency facilities and then finds the optimal strategies for distributing lifeboats. The results highlight the significant impact of overhead power line obstructions on the optimal rescue lifeboat distribution. Additionally, the results suggest prioritizing emergency evacuation for socially vulnerable households in Manville township. Practically, the generated rescue sequence and rescue lifeboat distribution are expected to help emergency response agencies perform effective and rapid rescue operations.

1. Introduction

Under climate change, the frequency and intensity of extreme precipitation events are increasing locally, especially in the northeastern United States [1]. Compared with coastal flooding induced by storm surges that can often be predicted days in advance, flooding caused by heavy precipitation can develop within minutes to hours, and the exact locations and severity are hard to predict, meaning shorter response time, more minor spatial scales, and longer lasting time [2,3]. Besides, precipitation-induced flooding could lead to inappropriate self-evacuation due to underestimations of hazards of shallow but speedy water flow [4,5], which places

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considerable demands and requests for efficient emergency response. Unlike rural environments, urban environments are typically featured by dense populations, extensive infrastructure, and intricate urban topography. These increase the complexity of emergency response efforts because the large volume of people needing evacuation can lead to traffic congestion and delays; the interdependent infrastructure systems can cause cascading failure that exacerbates flood impacts; and the varying ground elevations and land coverages can pose various levels of damage to different communities.

Transportation systems play an essential role in emergency evacuations and search and rescue. Flood impacts to transportation systems, such as eroded and undermined roadbeds, spawned debris, and high-water incidents, can lead to road closure that hinders emergency response and rescue operations. To address this issue, a vast amount of research has focused on identifying flood damage to transportation systems and analyzing traffic accessibility caused by the damage. For example, natural language processing and image processing techniques have been applied to extract flood damage-related information (e.g., damage severities and damage types) from publicly available image and text information [6,7]. Other research has modeled the topology of transportation systems to understand infrastructure cascading failures and identify vulnerable components through network analysis and simulation [8–10]. For a better understanding of residents' traveling patterns during flood evacuation, research studies have used Agent-Based Modeling (ABM) to simulate residents' evacuation to identify congested road sections and vulnerable areas [11,12] and utilized machine learning-based approaches to rapidly predict hurricane evacuation traffic flows [13]. Compared to obtaining information on road damage severity and road network accessibility, another line of work has focused on modeling the logistics of moving evacuees from flooded areas to safe and accessible places or shelters [14]. It has been found that private vehicle evacuation creates congestion and is also expensive and time-consuming [15,16]. As a result, emergency evacuations using public transit have attracted more attention. These studies generally involve developing optimization models to determine the number of public transits at each public accessible point given the constraints of evaluation time frame and public transit numbers [15–17]. Despite facilitating efficient emergency evacuation and rescue, existing studies are less applicable to rescue operations in flash flooding.

Flash flooding presents significant challenges for early evacuation and rescue operations due to its rapid water accumulation and localized severity, leaving little preparation time. Unlike coastal flooding, which often leaves time for advance warnings and allows for preemptive early evacuations, flash floods can occur within hours, trapping people in their homes and making self-evacuation difficult due to the seemingly shallow but speedy water flow. In such scenarios, lifeboats play a crucial role compared to cars because they are one of the few effective means of rescuing trapped individuals and transporting them to safety areas. However, lifeboat deployment faces challenges due to the complicated urban features. One of the most essential features is overhead powerlines. For instance, downed powerlines can cause electrocution and fire hazards, which makes it dangerous for evacuees and emergency responders to navigate the area. Fallen powerlines can obstruct roads and pathways, which limits routes for evacuation and causes traffic congestion and delays in reaching safe areas. During the emergency planning phase, ignoring the obstruction of overhead powerlines on emergency response activities could lead to biased decision-making [18], including unreliable estimations of road accessibilities, incorrect assumptions about the availability of rescue services, and unbalanced allocation of rescue resources.

Besides the availability of emergency services, social equity also plays a crucial role in emergency response activities because socially vulnerable households (e.g., elders and children) are often exposed to more significant risks and are less capable of safely evacuating by themselves. For example, elders usually face mobility challenges, chronic health conditions, or sensory impairments that limit their ability to evacuate independently or respond quickly to rising waters. Children, on the other hand, lack the physical strength, cognitive ability, and situational awareness to navigate or escape life-threatening conditions. Although the importance of focusing on vulnerable communities during emergency evacuation and rescue is widely known [18], existing studies that have considered social aspects are more oriented toward long-term flood mitigation, recovery, and rebuilding [19–21]. A few studies have considered social vulnerability during short-term emergency response [17]; however, these studies have mainly focused on impassible road sections caused by floods and have yet to integrate the obstruction caused by overhead powerlines [22–24].

To address these research challenges, this study aims to design a systematic approach to facilitate rapid and equitable emergency rescue during flash flooding in urban environments. Firstly, the present study employed an emergency response framework [18,25] to simulate the time-varied rescue routes during flooding expansion. Then, an integrated metric is designed to quantify rescue demands at the census block level. The metric is composed of three key components: rescue efficiency, which measures how fast emergency responders can travel from emergency facilities to the flooded residential buildings considering impassible road sections and overhead powerline obstruction; community flood severity, which quantifies flood damage to buildings; and social vulnerability, which is derived based on four household composition-related factors (elders, children, minority, and single parents households). Based on this metric, a systematic step is designed to suggest the sequence of rescue operations. Four lifeboat allocation strategies are designed to inform the optimal allocation of lifeboats at emergency facilities.

A case study in Manville, New Jersey, following the impacts of Hurricane Ida, was conducted to demonstrate the applicability and feasibility of the proposed approach. A sensitivity analysis was performed to evaluate how the weightings for the three factors (rescue efficiency, community flood severity, and social vulnerability) in the integrated metric influence lifeboat allocation and rescue efficiency. The finding suggests that a higher weight should be assigned to social vulnerability in Manville township, followed by community flood severity and rescue efficiency. Academically, the research designed a systematic approach to facilitate rapid and equitable emergency rescue using lifeboats during flash flooding in urban environments. This research innovatively considers the challenges associated with lifeboat deployment due to the complicated urban features, overhead power lines, and the higher rescue demands from socially vulnerable communities. In practice, the research provides decision support for efficient prioritization of rescue tasks and the optimal allocation of lifeboats during flash flooding events.

The remaining sections are organized as follows. The methodology section first describes the compositions of the integrated metric and then explains the mathematical details of the rescue demands quantification and the different types of rescue resource allocation

strategies. The case study section introduces the basic assumptions, data processing, and preliminary results. Finally, a discussion of the study findings, research contribution, and limitations are summarized.

2. Methodology

The methodology comprises four parts: data preparation, pre-processing, integrated metric calculation, and rescue lifeboat allocation strategies. In this section, the calculation of the integrated metric, which incorporates rescue efficiency, community flood severity, and social vulnerability, is first introduced to quantify rescue demands. Based on this metric, a systematic step is designed to suggest the sequence of rescue operations, and four types of strategies are designed for lifeboat allocations. An illustration of the research methodology is shown in Fig. 1.

2.1. Dynamic rescue demand quantification

Following flash flooding, as water levels gradually recede, the status of road accessibility changes because some of the road sections previously inaccessible by rescue boats or cars would become accessible. As a result, the time-varying characteristics of road accessibility would change and consequently impact the rescue demands. This study designs an integrated metric to quantify rescue demands at various time steps for rescue operation prioritization. This metric is composed of three key factors. First is rescue efficiency, which indicates the ability of responders to travel from emergency facilities to impacted households. This is critical because it helps responders understand community accessibility, allowing lifeboats to be allocated effectively and rescue operations to be conducted efficiently. Second, community flood severity quantifies the extent to which flood impacts destroy the buildings. Considering this factor is essential because flooded or structurally compromised buildings can collapse, posing immediate threats to residents. The selection of these two factors is also supported by the study of Feng et al. [26], which highlighted that areas with higher inundation rates and more severe flood conditions tend to have a higher percentage of people seeking rescue. Third, social vulnerability identifies the households that require more significant assistance during emergency rescue. In detail, socially vulnerable populations, such as the elderly, children, and individuals with disabilities, often face more considerable challenges in self-evacuation and require additional assistance during rescues. A detailed explanation of each of these parts is given below.

2.1.1. Rescue efficiency

Rescue efficiency is defined as how fast emergency responders can travel from their emergency facility (i.e., fire stations and emergency operation centers) to the flooded residential buildings. To consider the dynamic characteristics of road accessibility, an attributed road network is modeled to calculate rescue efficiency. Specifically, topological road attributes (road sections and road intersections) and building locations (i.e., emergency facilities and residential houses) are used to create an attributed network in which the nodes represent locations of emergency facilities, flooded residential buildings, road sections, and road intersections, and edges represent connections between road sections and connections between buildings and road sections. Three attributes are assigned to road sections and intersections: accessible by cars, accessible by boats, and not accessible by boats. These are dynamically updated, considering the changes in flood depth and overhead powerline obstruction. If a road section/intersection becomes flooded and inaccessible due to overhead powerline obstruction, this node and its adjacent edges are removed from the network. An illustration of the dynamic process is illustrated in Fig. 2. As the flood recedes, the status of the highlighted road section changes (indicated in orange) from not accessible by boats to accessible by lifeboats and accessible by cars. The bottom part shows the disrupted road networks.

The rescue efficiency (RE) is quantified as the inverse of the shortest path from an emergency facility to a flooded residential building. Mathematically, given a disrupted network G_t at timestep t , RE is formulated in Equation (1).

$$RE(G_t) = \frac{1}{d_{qt}} \quad t \in 1, 2, \dots, T \quad (1)$$

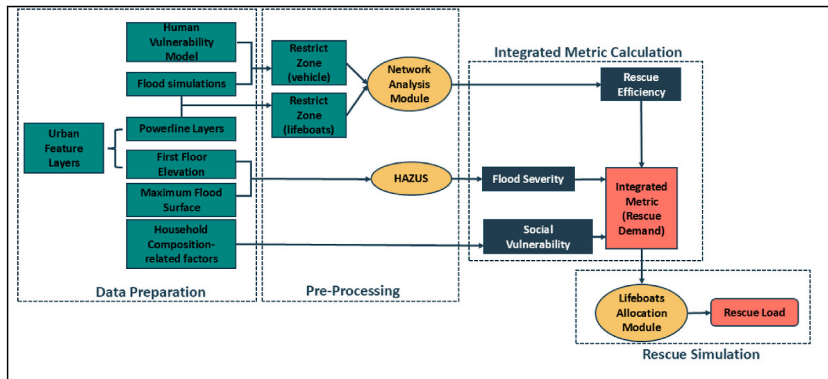


Fig. 1. Research methodology flow chart.

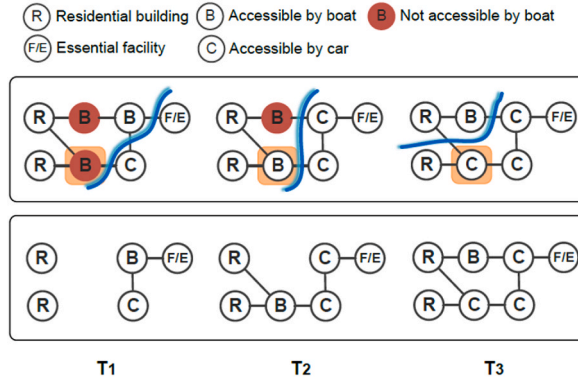


Fig. 2. Illustration of the changes in road networks over time.

Where q represents the emergency facility, n represents the flooded residential building and d represents the shortest path from the emergency facility to the flooded residential buildings. The greater the rescue efficiency, the faster the emergency responders can reach the flooded building.

2.1.2. Community flood severity

During emergency rescue, the extent to which a building is damaged influences rescue demands. The reason is that buildings more damaged by floods are at a higher risk of collapse. This poses an immediate threat to residents still living inside, making it urgent to rescue them before further deterioration occurs. Considering this effect, community flood severity is quantified as the percentage of flood-induced building damage. Due to the unique characteristics of building design attributes (e.g., the presence of a basement, first-floor elevation, and construction materials), the extent of building damage would vary significantly even with the same flood elevation. To derive the percentage of building damage, HAZUS—a GIS-based software developed by the Federal Emergency Management Agency (FEMA) to identify areas with high risk for natural hazards and estimate the physical, economic, and social impacts of hazards—is employed [27]. HAZUS employs depth-damage curves, which relate the depth of flooding to the percentage of damage for different types of buildings and are derived based on empirical data and engineering studies. The greater the percentage of building damage (PB), the more urgent it is to rescue residents living inside these buildings. It is worth noting that the depth-damage curve can also be manually specified using historical data. This often led to a more accurate building damage estimation. However, when such data are unavailable, HAZUS can be used as a convenient tool for general analysis. Since a building is less likely to become restored during flood impacts, the percentage of building damage is computed based on the maximum simulated flood depth and is not affected by time.

2.1.3. Social vulnerability

Focusing on vulnerable households during emergency evacuation and rescue is critical because these groups are often exposed to more significant risks and face more barriers to evacuating to safety. In this study, four household composition-related factors are selected: the percentage of persons aged 65 and older ($P_{65older}$), the percentage of persons aged 18 and younger ($P_{18younger}$), the percentage of minority households ($P_{minority}$), and the percentage of single-family households ($P_{singlefamily}$). Due to privacy concerns, identifying these four factors at the building level is not possible. As an alternative, census block data, the smallest unit of public data collected by the U.S. Census Bureau can be used [28]. A percentile ranking approach is used to integrate the four factors. Specifically, each of the four variables is first ranked from the highest to lowest across all census blocks in the investigated community, where a higher ranking indicates a more vulnerable census block. Then, a percentile rank was calculated for each census block over these variables. In the end, the percentile ranks were summed to indicate a block's evacuation vulnerability SV_k . By prioritizing census blocks that have higher rankings, emergency responders can ensure that vulnerable communities are not disproportionately affected by emergencies and that everyone has the opportunity to evacuate safely and receive the necessary support.

2.1.4. Rescue demands

The rescue demand is defined as a metric that integrates rescue efficiency, community flood severity, and social vulnerability, as introduced above, and it is used to inform emergency rescue at the census block level. Since rescue efficiency and community flood severity are derived at the building level, these metrics are first averaged at the blocked level. For a census block k with n number of residential buildings, the block-level rescue efficiency ($RE_{k,t}$) at a time step t is shown as follows:

$$RE_{k,t} = \frac{\sum_{N=1}^n \frac{1}{d_{qn_t}}}{n} \quad k \in 1, 2, \dots, m, t \in 1, 2, \dots, T \quad (2)$$

The block-level flood severity is calculated as follows:

$$FS_k = \frac{\sum_{n=1}^n PB_n}{n} \quad k \in 1, 2, \dots, m \quad (3)$$

Given the block-level rescue efficiency $RE_{k,t}$, flood severity FS_k , and social vulnerability SV_k , the rescue demands of a census block k at timestep t is ($RD_{k,t}$) computed by first calculating the percentile rank of each variable across all census blocks in an investigated community. Here, flood severity and social vulnerability are ranked from the highest to the lowest because a higher value indicates a greater demand for rescue. On the contrary, rescue efficiency is ranked from the lowest to highest because a higher efficiency indicates better rescue ability. Based on the percentile ranks, these factors are aggregated by a weighted sum. In detail, each factor is assigned to a specific weight, where the weights are non-negative and sum up to 1. The percentile ranks of each factor are multiplied by its corresponding weight, and the resulting products are summed to compute the final $RD_{k,t}$. Mathematically, $RD_{k,t}$ is calculated as follows:

$$RD_{k,t} = w_1 * RE_{k,t} + w_2 * FS_k + w_3 * SV_k \quad (4)$$

where $w_1 + w_2 + w_3 = 1$ and $0 \leq w_i \leq 1$ for $i = 1, 2, 3$. In the end, a percentile ranking based on the integrated metric $RD_{k,t}$ is derived to inform emergency rescue prioritization. In practice, these weights could be determined based on community characteristics. For example, a community with more elders and children could assign a higher weight to the social vulnerability factor, and a community with poor-quality houses could assign a higher weight to the community flood severity factor.

2.2. Rescue lifeboat allocations

Lifeboats are commonly used for rescue in inundated areas during flooding and are the resources mainly concerned in this study. To evaluate how the obstruction of overhead powerlines and the consideration of socially vulnerable households impact the efficiency of emergency response planning and execution, we first calculated the association between census blocks and available emergency facilities based on network analysis at dry conditions. Specifically, each building was matched with the optimal emergency facility offering the shortest path (i.e., highest efficiency). For each census block, the most frequently associated emergency facility among individual buildings was identified as the associated facility for that block. Additionally, four different lifeboat allocation strategies are designed for comparison. First, the ‘equal plan’, which evenly distributes lifeboats among available emergency response facilities. The second one is named ‘Plan 1’. It allocates lifeboats based on the ratio of the number of accessible census blocks needing rescue associated with each emergency facility. Needing rescue means a specific census block was flooded at least once during the development of flooding. This means a facility with a higher count of accessible census blocks that potentially have rescue requests will be assigned with more lifeboats. This strategy requires a pre-understanding of the accessibility of buildings and blocks. ‘Plan 2’ follows a similar pattern as ‘Plan 1’ but considers the total number of associated blocks per facility regardless of the rescue requirements. ‘Plan 3’ uses the integrated metric, rescue demands, to inform lifeboat allocation. Similarly, a ratio is specified. The difference is that the ratio is calculated based on the summation of the integrated metrics of all associated census blocks with rescue requests for each facility. A higher summation indicates that the associated accessible census blocks have higher vulnerability and longer travel distances to the facilities, signifying a greater need for lifeboats.

The rescue operations were simulated based on the assumption that a single lifeboat would be used to rescue all residential

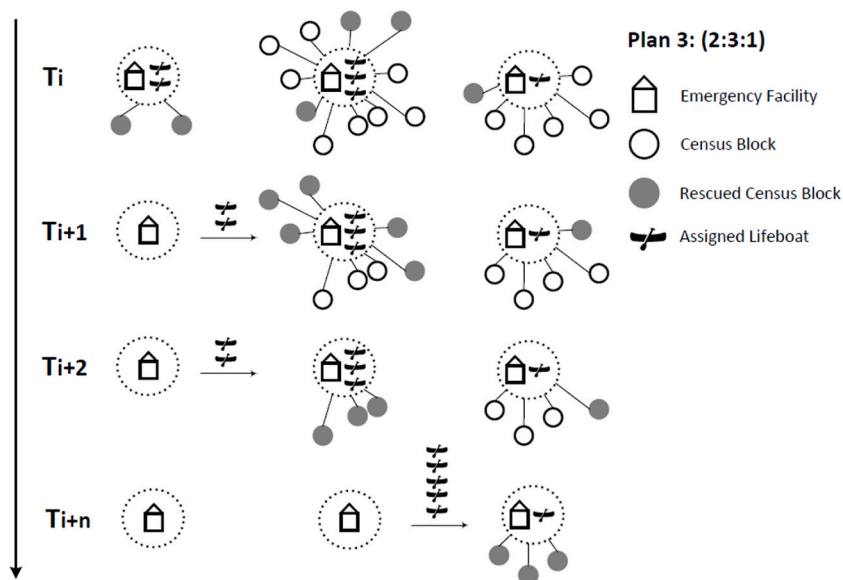


Fig. 3. Illustration of the dynamic allocation process of Plan 3.

buildings within a specific census block at one time. This assumption is based on two key considerations. First, early evacuations have already taken place, meaning that only a small number of individuals remain stranded by floodwater within each census block. Second, within the same census block, we assume that buildings are mutually accessible using lifeboats. Consequently, if at least one building in a census block is accessible, the entire block is considered accessible. To better represent rescue operations, the allocation ratio of lifeboats for each strategy dynamically changes based on the latest rescue progress. Assuming six rescue boats are available, four allocation ratios (based on the four defined strategies) are assigned based on the flood conditions at the start time of rescue. Once the number of associated blocks for a specific emergency facility reaches zero, the assigned lifeboats will be released and reallocated to other facilities according to the assignment logic of each strategy. An illustration of Plan 3 is given in Fig. 3 and the illustrations of different plans are provided in Appendix C, Figure C.1 – C.3.

To track rescue progress, two status flags are introduced at each time step: the inherit flag N_1 and the rescue flag N_2 . Initially, for all census blocks that will be flooded, N_1 is set to 1, indicating that none of these blocks have been rescued. At every subsequent time step, N_1 is updated to 0 for blocks that have been rescued and this status is inherited to the next time step. The rescue flag N_2 reflects the accessibility and flood hazards of specific census blocks; it is set to 1 for blocks that will be flooded and are accessible at the current time step and 0 otherwise. At each time step, a new metric $CM_{k,t}$, calculated based on Equation (5) for each census block. It determines the rescue necessity, feasibility, and emergency of all census blocks at specific time step t . Then, by ranking the metric, the census blocks with the top-ranked metric will be rescued at time step t .

$$CM_{k,t} = N_{1,k,t} \cdot N_{2,k,t} \cdot RD_{k,t} \quad k \in 1, 2, \dots, m, t \in 1, 2, \dots, T \quad (5)$$

Where m is the number of impacted census blocks. The rescue load RL , defined as the sum of the rescue demands when the rescue operations start, of all remaining census blocks with rescue request ions during flooding (Equation (6)), will be employed here to track the rescue progress using different allocation strategies. In practice, the rescue load represents the total vulnerability of all remaining census blocks with potential rescue requests at specific time steps.

$$RL_t = \sum_k^m N_{1,k,t} \cdot N_{2,k,t} \cdot RD_{k,t_0}, k \in 1, 2, \dots, m, t \in 1, 2, \dots, T \quad (6)$$

Where t_0 is the time when the rescue operations start, T is the number of time steps.

3. Case study

The flash flooding in Manville, NJ during Hurricane Ida was selected to demonstrate the feasibility and applicability of the proposed approach. Officials report that fluvial or river floodings are prevalent throughout Manville in areas proximate to the Raritan River and the Millstone River [29]. It has faced repeated flooding events, notably during Hurricane Floyd in 1999, Hurricane Irene in 2011, and the remnants of Hurricane Ida in 2021. These floods have caused significant property damage, displaced residents, and strained emergency response systems. Manville is a socially vulnerable township. The median household income is approximately \$80,000, with a poverty rate of nearly 10 %. In the township, about 18.4 % of the population is under the age of 18, and about 18.25 % of the total population is elderly population, those aged 65 and older [30]. These statistics highlight the significant proportion of vulnerable groups, including children, senior citizens, and minorities, who are less capable of self-evacuate and require prioritized assistance during emergency rescue. From September 1st through September 3rd, 2021, Tropical Storm Ida moved through New Jersey, causing high winds and heavy rainfall. The Raritan River crested at about 27.6 feet, the highest ever recorded (the previous highest being Hurricane Floyd with a crest of 27.1 feet in 1999) [31]. During Ida, water levels in some areas of Manville reached up to 6 feet, rendering many roads unusable and making it difficult for residents to evacuate. In some regions, entire neighborhoods were cut off, making it impossible for residents to leave safely due to impassable roads and rising waters. These complicate rescue operations as only boats could be used once water levels exceeded 2–3 feet [18]. In some cases, overhead power lines pose significant challenges to boat-based rescues, as live lines can electrify floodwaters, endangering rescuers and evacuees, while sagging or damaged lines may physically obstruct boat movement. Given these challenges, it is critical to quantify building damage reliably, comprehensively evaluate community accessibility, prioritize the needs of the socially vulnerable population, and effectively allocate rescue boats for efficient rescue operations.

This section first introduces the details of Manville's flood condition reconstruction and overhead powerline extraction. Then, steps taken to calculate the dynamic rescue demands are explained. After this, the rescue progresses after the proposed lifeboat allocation strategies are compared, and the reasons for different patterns are discussed.

3.1. Data processing

3.1.1. Flood reconstruction

The flood conditions for Hurricane Ida in this study were reconstructed by a street-scaled 2-dimensional hydrodynamic model [32], validated by measured high water marks. The model domain encompasses the Raritan River Basin area from Branchburg to Bound Brook Township, incorporating three major upstream freshwater inputs. The river bathymetry, integrated into the background terrain used by the model, has been enhanced based on the most recent bathymetry dataset created by Rutgers University [33]. Additionally, to accurately capture the impact of buildings on flood spreading, building footprints were converted to a raster dataset with a

consistent building height of 10 m and then merged onto the background terrain. A spatially varied resolution mesh was employed, featuring a 3-m resolution in flood-prone street areas and a resolution ranging from 10 to 50 m for the remainder of the domain. The original shallow water equations, the Eulerian-Lagrangian Method (SWE-ELM), were employed for performing two-dimensional unsteady flow routing in this case study.

3.1.2. Overhead powerline extraction

The methodology for extracting overhead power lines from a citywide point cloud dataset involves several key steps to ensure accurate and useful data. Initially, a point cloud dataset was collected, which includes detailed spatial information of the target area. This dataset is rich with various urban features such as buildings, trees, and overhead power lines. The extraction process begins by digitizing these overhead power lines into 3D polylines using specialized software like VRMesh V11.8.1 and ArcGIS 10.8.1. In VRMesh, the point cloud data was processed to identify and highlight the power lines. This involves semi-automated procedures to extract the elevation and location information of the power lines. However, due to potential spatial discontinuities caused by obstructions like vegetation, the extracted data may be incomplete. To address this, spatial interpolation techniques in ArcGIS were applied. The digitized polylines were converted into polygons with a buffer distance to facilitate accurate interpolation. This process ensures a continuous and coherent dataset representing the overhead power lines. Finally, the elevation and location information of the extracted power lines were exported as raster layers to assess the clearance distances above potential floodwater levels at each time step, ensuring the safety and effectiveness of rescue operations during flood events [18].

3.1.3. Rescue efficiency

Based on the road network (polylines) and building footprints (polygons) in Manville, a citywide network was generated using a series of ArcPy scripts to connect all residents and emergency facilities (Fig. 3). Road nodes were created every 13.7 m (45 feet) along the road segments and assigned the type “C” or “B” to represent the accessibility status of each node. “C” indicates accessible by vehicles, while “B” indicates accessible by lifeboats. All building footprints within the study area were converted to building nodes and assigned the type “R,” “E,” or “F,” “R,” represents normal residential and commercial buildings, “E” represents evacuation centers, and “F” represents fire stations, with both “E” and “F” nodes being emergency response facilities. Each building node was then linked to its nearest road node, representing the connection status between the two nodes in the network.

At each time step, the accessibility status of all nodes was adjusted based on flood progression. For example, road nodes in vulnerable areas, identified using the human instability model [5,34], will change from “C” to “B,” indicating that these nodes are no longer accessible by vehicles and pedestrians. The assumption here is that areas that cannot be accessed by walking also cannot be accessed by vehicles due to higher relative motion speeds and limited flexibility in floodwaters. Additionally, the clearance distance between the water elevation and overhead power lines was calculated at each time step and compared with the standard clearance distance for safety considerations to identify restricted zones for lifeboats due to overhead power line obstructions [18]. In this case study, we update the obstruction layer every 1 h. Paths within these restricted zones were removed at each step to indicate inaccessibility. As shown in Fig. 4, on September 2nd, 2021 at 12:00 a.m., all road nodes in walking-restricted areas changed from “C” to “B,” and all paths covered by boat-restricted areas were removed. Three fire stations and one evacuation center were identified as

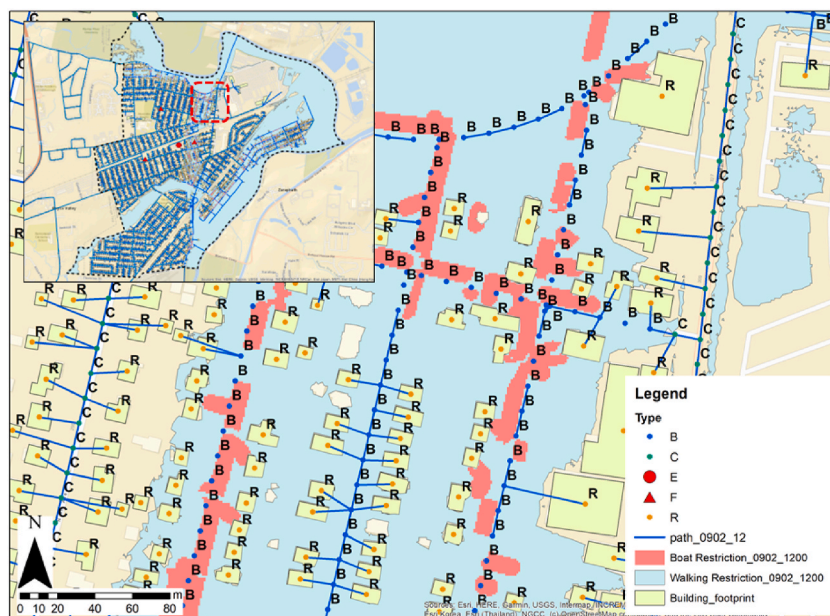


Fig. 4. Example network accessibility on September 2nd, 2021 at 12:00 a.m.

facilities that can provide emergency rescue. However, one of the fire stations was flooded during Hurricane Ida and lost its ability to provide emergency response. As a result, it was excluded from the lifeboat allocation and rescue simulation. In total, three essential facilities are identified in Manville, and the rescue efficiency is determined based on the closest accessible facilities.

Given the generated network at each timestep, the rescue efficiency for each census block at each hour is calculated following Equation (2).

3.1.4. Community flood severity

HAZUS was utilized to estimate the damage percentage to individual buildings using the processed building inventory data and the maximum water depth obtained from the flood reconstruction step (section 3.1.1). Here the building inventory data, such as the number of stories, roof type, exterior wall covering, and the presence of a basement, was processed by gathering data from njparcels.com and njpropertyrecords.com and surveying all the homes using Google Street View (GSV). One of the most important building attributes, first-floor elevation, a critical factor in determining a building's risk, was extracted from mobile LiDAR data collected by the Rutgers mobile mapping team following Hurricane Ida. Innovative technology, using Yolo5, a 2D object detector for identifying building components like windows, doors, and garage doors based on the intensity of the point cloud, is employed to extract the first-floor elevation [35]. Given the building damage percentage, Equation (3) calculates community flood severity at each census block. An illustration of the building damage percentage for buildings with damage percentages greater than zero is indicated in Fig. 5. The graduated colors changing from white to blue represent the percentage of building damage, where a whiter color shows a low percentage of damage and darker blue shows a high percentage of damage.

3.1.5. Social vulnerability and integrated metric

The block-level house composition data (i.e., the percentage of persons aged 65 and older, people aged 18 and younger in minority households, and single-family households) is obtained from the U.S. Census Bureau [23]. Following the proposed integrated metric calculation approach, the rescue demands of each census block at each hour were calculated. An illustration of the details of the metric calculation process at one timestep is shown in Appendix A, Table A.1. An example of the spatial distribution of rescue demands and its three components is shown in Fig. 6.

Ultimately, a percentile ranking is derived based on the integrated metric to inform emergency rescue prioritization. The percentile ranks are dynamically updated at each time step based on the latest road conditions.

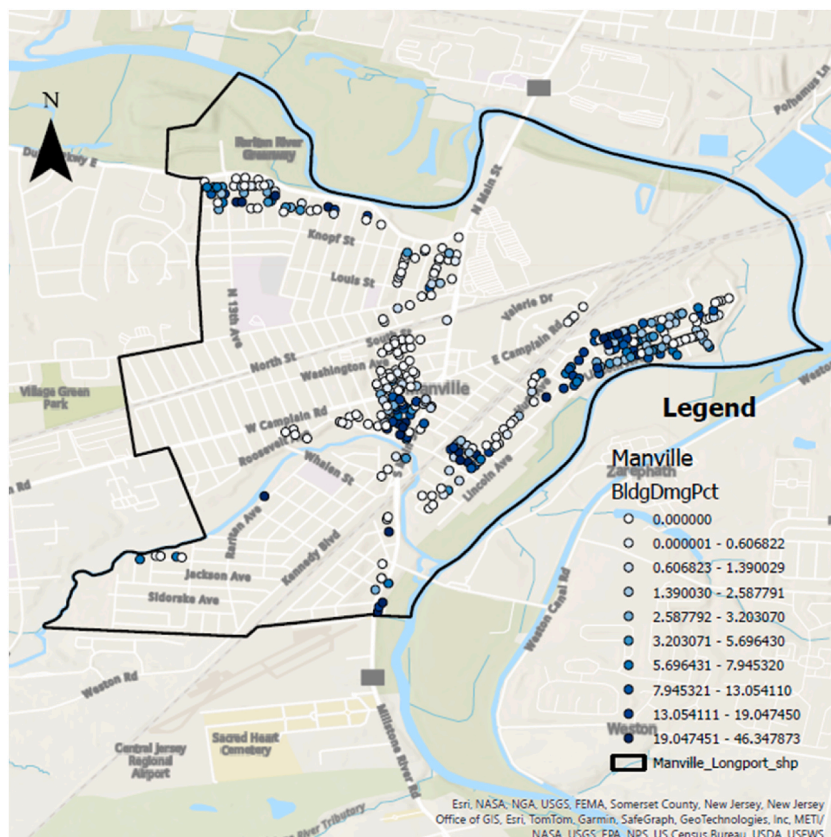


Fig. 5. Percentage of building damage at Manville.

3.2. Results

To compare and fully understand the performance of different allocation strategies, the rescue progresses under two scenarios are evaluated. These are considering and neglecting the obstruction of overhead powerlines. Assuming the total available lifeboats for rescue operations is 6, the initial allocation ratio for the four strategies is [2,2,2] for the Equal Plan [1–3], for Plan 1 [1–3], for Plan 2, and [1–3] for Plan 3. These allocation ratios vary over time, as described in section 2.2.

In this study, rescue operations are set to begin at the 13th time step (September 2nd, 2021, at 12:00 a.m.), 4 h after the peak level detected in the nearest river gage, initially under complete rescue requirements. This assumption represents the worst-case scenario because the rescue loads are the highest [14]. When the three components of the integrated metrics are equally weighted, the changes in rescue loads under different scenarios with different allocation strategies are illustrated in Fig. 7. To better assess and compare the rescue load variation under various scenarios with different allocation strategies, the Area Under Curves (AUC) is calculated for all rescue-load lines (Table 1) to quantify the efficiency of rescue progress. A smaller AUC represents higher overall rescue efficiency, indicated by a rapid drop in rescue loads, and vice versa. In Table 1, the significant differences in AUC between scenarios considering overhead power lines and neglecting overhead power lines are from the reality that large amounts of census blocks are inaccessible due to the obstruction by overhead power lines with the spread of flooding. The number of rescuable census blocks would be much lower than expected if overhead power lines were neglected. In the cases of neglecting overhead power lines, the results illustrated that the lifeboat allocation considering social vulnerability (Plan 3-NP) is the most effective strategy (see Fig. 7), as it achieves the largest decrease rate in rescue load and also the smallest AUC (Table 1). Considering the obstruction of the overhead power lines, Plan 3-WP also returns the smallest AUC, representing the highest overall rescue efficiency. Plan 1, which considers potential rescue requests but does not account for social vulnerability, achieves the second-smallest AUC in both scenarios. On the other hand, Plan Equal, though a “sightless” strategy, outperforms Plan 2 in this case study. This outcome is unsurprising since both strategies are less targeted. In particular, Plan 2 focuses solely on associations while neglecting potential rescue requests. This neglect creates an imbalance between the total number of associated blocks and the number of blocks with actual rescue requests, resulting in misaligned allocations and further reducing its effectiveness. In scenarios considering the obstruction caused by powerlines during rescue activities, the rescue load decreases before rescue activities, as more blocks become inaccessible for lifeboats due to the proximity of rising floodwaters to the overhead powerlines. The increased rescue load after September 2nd at 23:00 is due to more blocks becoming accessible again as the water level recedes.

Results show that an efficient allocation strategy should consider the rescue requirement and accessibility status because the distribution of rescue demands among emergency facilities is uneven in reality. An allocation strategy without considering this uneven distribution will result in low effective rescue operations (Plan 2). For example, in this case study, facilities have a high number of associated blocks, but most of the associated census blocks are not impacted by flooding. Similarly, some facilities with fewer associated blocks have a relatively higher rescue demand because most associated census blocks are located in flood-prone areas. Plan 3 is a strategy that allocates lifeboats based on the sum of integrated metrics of all associated census blocks to each emergency response facility. The facility will be assigned more lifeboats if the associated buildings are more vulnerable (higher sum of integrated metric).

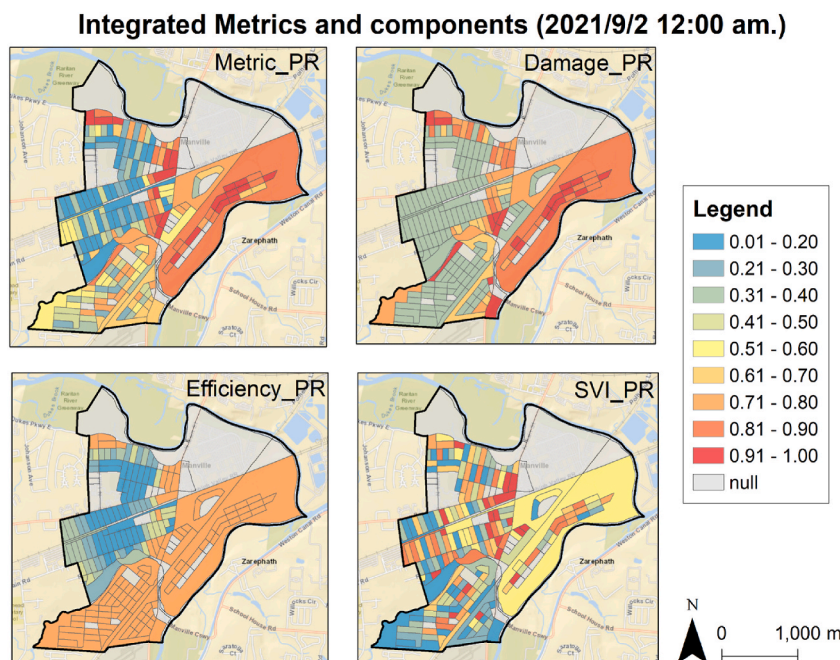


Fig. 6. The spatial distribution of rescue demands and its three components (equally weighted) on September 2nd, 2021, at 12:00 a.m.

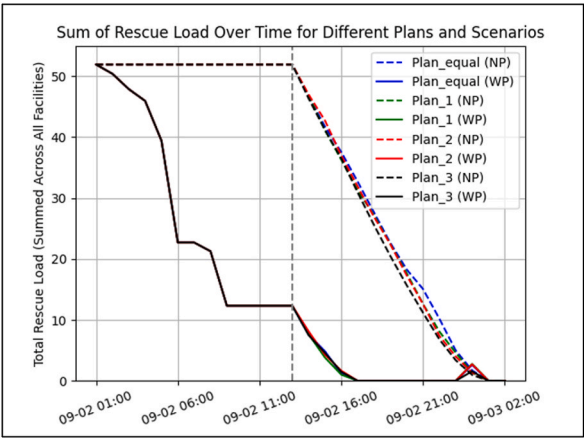


Fig. 7. The rescue load variation using different lifeboat allocation strategies under two scenarios, rescue operations considering powerline obstruction (solid lines) or neglecting powerline obstruction (dash lines). The left part of the gray dash line indicates the rescue load before rescue activities. NP refers to ‘No Powerlines Consideration Scenario’, and WP refers to ‘With Powerlines Consideration Scenario’.

Table 1
Comparison of AUC among scenarios and different allocation strategies.

Area Under Curve (AUC)	WP	NP
Plan Equal	0.139	0.422
Plan 1	0.134	0.408
Plan 2	0.143	0.410
Plan 3	0.133	0.393

Compared with Plan 1, Plan 3 provides a strategy that considers more comprehensive features when implementing rescue operations and also outperforms the other three strategies, illustrating the advantage of considering social vulnerability in rescue operations.

4. Discussion

In practice, the weighing of the rescue efficiency, community flood severity, and social vulnerability would differ due to the prioritization needs of various communities. A sensitivity analysis was conducted to evaluate the influence of weights for different factors on rescue operations. In the sensitivity analysis, the weights for each of the three factors varied between 0.1 and 1, in increments of 0.1, with the total weight constrained to 1. This resulted in 36 possible combinations with non-zero weights and multiples of 0.1. The detailed weight combinations and the corresponding case ID are shown in Appendix B, Table B.1. For each possible combination, the four lifeboat allocation strategies (with and without overhead powerlines) were followed to generate the changes in rescue load over time. Comparatively, the efficiency of rescue operations that neglect overhead power lines (shown in Appendix B, Figure B.1) is significantly lower than the allocation strategies that consider overhead power lines (Fig. 8).

In Fig. 8, the x-axis represents the weight combinations, with WID 0 denoting the equally weighted case and WID 1 to WID 36 corresponding to the 36 unique weight combinations. The y-axis represents the four lifeboat allocation strategies, dyne, dyn1, dyn2 and dyn3 represent Plan Equal, Plan 1, Plan 2 and Plan 3, respectively. The number and color in each grid show the AUC value for a specific weight combination and allocation plan. The figure shows that Plan 3 (dyn3) consistently achieves the smallest AUC for most weight combinations, indicating better overall rescue efficiency. The main difference between Plan 3 and the other plans is the consideration of social vulnerability. This result highlights the importance of incorporating social vulnerability as a critical factor in rescue operations planning.

Focusing on plan 3, a ternary plot was generated to demonstrate the changes in AUC values under 37 wt combinations (Fig. 9). In Fig. 9, the color and sizes illustrate the values of AUC. The top three weight combinations, in the order of (community rescue

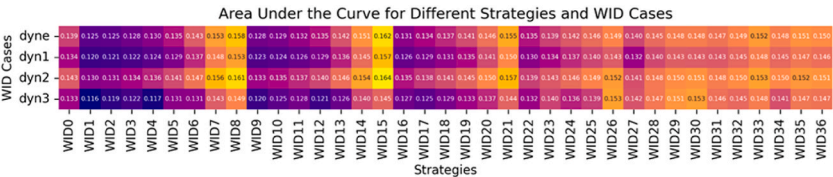


Fig. 8. Area Under the Curve for different strategies when considering overhead power lines with different weight combinations.

efficiency, flood severity, and social vulnerability), are identified as (0.1, 0.1, 0.8), (0.1, 0.4, 0.5), and (0.1, 0.2, 0.7). These combinations are recommended for rescue operations planning in Manville under the conditions of the Ida flood scenario.

The changes in AUC values with varying factor weights are visualized and compared to further evaluate how rescue efficiency, flood severity, and social vulnerability influence overall rescue efficiency (Fig. 10).

A decreasing trend in AUC (indicating improved overall rescue efficiency) is observed as more weight is assigned to social vulnerability. Conversely, increasing AUC values (reduced overall rescue efficiency) are observed when greater weights are assigned to rescue efficiency or flood severity. These findings suggest that prioritizing communities with higher social vulnerability leads to improved overall rescue efficiency in Manville. Notably, smaller AUCs (higher overall rescue efficiency) are achieved when weights assigned to social vulnerability are ≥ 0.4 , while those assigned to flood severity and rescue efficiency are < 0.4 . Comparing rescue efficiency and community flood severity, given the same weight (smaller than 0.4), lower AUCs were observed for flood severity compared to rescue efficiency. This finding indicates that prioritizing rescue operations for communities that experience greater flood damage contributes more to the overall efficiency of rescue operations than prioritizing communities with lower rescue efficiency.

It is worth noting that the observations made in the case study only apply to Manville during rescue operations under the impacts of flash flooding. The proposed approach has two disaster-related factors: community flood severity, which is calculated based on the percentage of flood damage to buildings, and rescue efficiency, which is calculated considering flooded road sections and overhead powerline obstructions. A more systematic investigation is needed to generalize this approach to other urban flood scenarios or different types of disasters. This includes examining the building damage mechanisms specific to various disasters to derive community damage severity and identifying key factors that hinder emergency rescue to determine rescue efficiency.

5. Limitation

There are several challenges to implementing the proposed approach in practical rescue scenarios, primarily due to the simplified assumptions about the rescue process and the exclusion of unanticipated disruptions during emergency rescue operations. The following section provides a detailed analysis of these limitations and their implications for emergency evacuation and rescue.

For general planning purposes, this study simplifies the rescue process by focusing on the accessibility of impacted census blocks. If a census block is deemed accessible, all households within it are assumed to be evacuated, and the rescue loads decrease by the rescue demand (integrated metric) associated with that block in one timestep. However, practical rescue operations involve additional complexities. In detail, constraints such as the number of impacted residents, lifeboats' capacity, and emergency facilities' capacity significantly influence the rescue process. When these factors are considered, the assumption that all affected households in an accessible census block can be rescued simultaneously becomes invalid. Instead, only a portion of the flooded households may be evacuated at a time, resulting in a smaller decrease in rescue load per timestep.

In real-world scenarios, unanticipated flood disruptions on infrastructure systems, e.g., fallen powerlines collapsed bridges, and inundated tunnels, can pose significant challenges during the rescue evacuation. Specifically, once a rescue operation is planned, these disruptions may alter the community's flood severity and affect rescue efficiency during the execution phase. This requires real-time identification of infrastructure failure [24,36] and necessitates recalculating key factors and adjusting the associated weights to reflect the updated conditions. While such recalculations can be performed quickly using automated procedures, effectively reallocating previously assigned lifeboats remains a logistical challenge. Real-time coordination and resource reallocation under dynamic and uncertain conditions are critical yet complex aspects that could influence the overall success of the rescue efforts [37]. Additionally, methods supporting real-time identification of infrastructure damage are also desired.

6. Conclusion

The unpredictable and localized nature of precipitation-induced flooding and the obstacles resulting from infrastructure damage and high-speed water flow make emergency evacuation and rescue challenging. To help emergency responders carry out effective and rapid rescue operations, this study designs a systematic approach to facilitate fast and equitable emergency rescue during flash flooding in urban environments. An integrated metric is designed to quantify rescue demands at the census block level. The integrated metric considers rescue efficiency, community flood severity, and social vulnerability. These are defined as the ability of responders to travel from emergency facilities to impacted households, the extent to which flood impacts destroy the buildings, and the households that need more emergency support. Following this, a systematic step is designed to suggest the sequence of rescue operation prioritization and four types of lifeboat allocation strategies are designed to inform the optimal allocation of lifeboats at emergency facilities.

To demonstrate the applicability and feasibility of the proposed approach, a case study in Manville, New Jersey, following the impacts of Hurricane Ida, was conducted. Using the designed integrated metric and the resource allocation strategies, a comparison of the efficiency of emergency rescue between two scenarios, considering or neglecting the impacts of overhead powerlines on lifeboat-based rescue operations during extreme flooding, was constructed to evaluate the effects of overhead powerline obstruction on emergency rescues. The results show that the strategy considering overhead power line obstruction returns the highest efficiency in all four resource allocation strategies. Among the four resource allocation strategies, the resource allocation strategy based on the proposed metric demonstrates the best performance, verifying the applicability of the designed metric. Furthermore, a sensitivity analysis was performed to evaluate how the weightings for the three factors (rescue efficiency, community flood severity, and social vulnerability) in the integrated metric influence lifeboat allocation and the overall rescue efficiency. The finding suggests that a higher weight should be assigned to social vulnerability in Manville township, followed by community flood severity and rescue efficiency.

Academically, the research designed a systematic approach to facilitate rapid and equitable emergency rescue using lifeboats

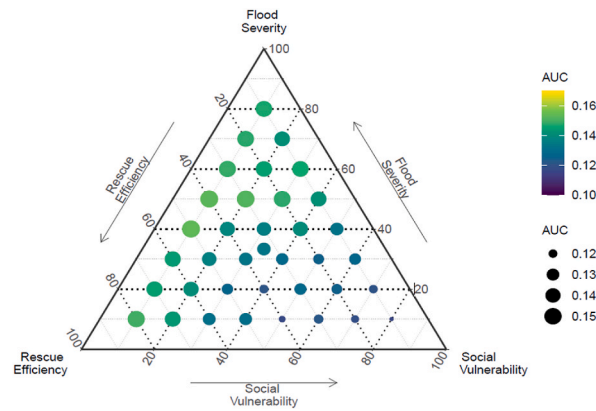


Fig. 9. Ternary plot illustrating changes of AUC values under 37 wt combinations.

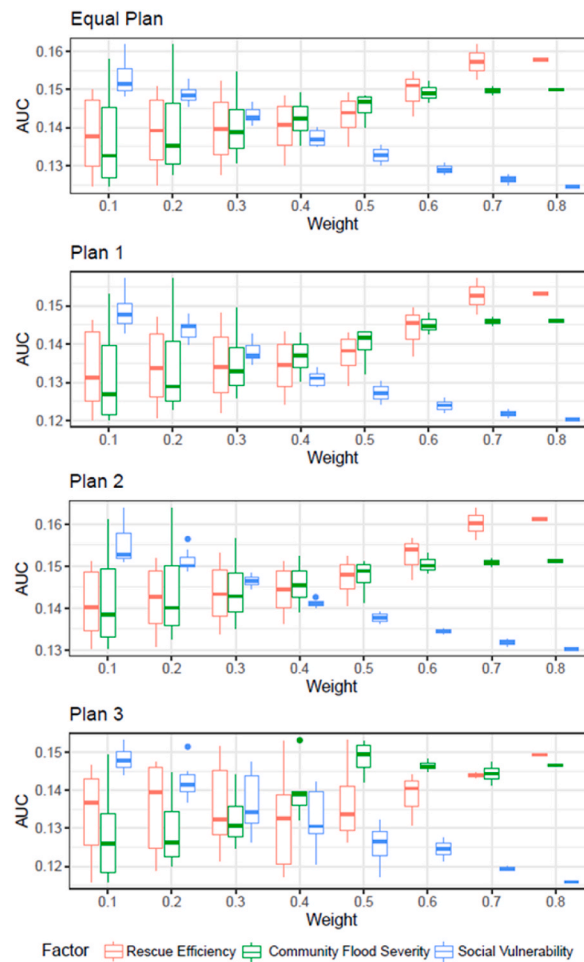


Fig. 10. Influence of AUCs corresponding to each resource allocation strategy with different factor weights.

during flash flooding in urban environments. This research innovatively considers the challenges associated with lifeboat deployment due to the complicated urban features (overhead powerlines) and the rescue demands from socially vulnerable communities. In practice, the research provides decision support for efficient prioritization of rescue tasks and the optimal allocation of lifeboats during flash flooding events. However, the current study has several limitations. First, the analysis ignores the constraints, such as the number of impacted residents, lifeboat capacity, and emergency facilities' capacity during the rescue evacuation process. Additionally, the

unanticipated flood disruptions on infrastructure systems, such as fallen powerlines, collapsed bridges, and inundated tunnels, may alter the community's flood severity and affect rescue efficiency during the execution phase. Future studies will incorporate these aspects to model real-life emergency evacuation processes by utilizing actual data collected during emergency response during Hurricane Ida at Manville. Future research will also investigate algorithms to perform real-time identification of the damaged infrastructure systems and update the rescue progress accordingly.

CRedit authorship contribution statement

Yitong Li: Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis. **Yifan Wang:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis. **Jie Gong:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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APPENDIX A. Integrated metrics calculation illustration

Table A.1

An illustration of the details of metric calculation process at one time step.

Block ID	Building Damage Percentage	Efficiency	Building Damage Percentage Percentile	Efficiency Percentile	SVI Percentile	Integrated Metric	Integrated Metric Percentile
1019514	0.095	0.007	0.943	0.912	0.866	0.907	1.000
1003514	0.110	0.009	0.964	0.809	0.892	0.888	0.995
1005514	0.097	0.009	0.948	0.825	0.887	0.887	0.990
1010514	0.052	0.006	0.907	1.000	0.716	0.875	0.985
1017515	0.000	0.056	0.327	0.052	0.196	0.192	0.021
2034516	0.000	0.041	0.327	0.108	0.093	0.176	0.015
2027516	0.000	0.100	0.327	0.010	0.098	0.145	0.010

APPENDIX B. Data and results in sensitivity analysis

A sensitivity analysis was conducted to evaluate the influence of weights for different components (rescue efficiency, flood severity and social vulnerability) of the integrated metrics on rescue operations.

Table B.1

wt combinations and the corresponding case ID

WID	w1	w2	w3	WID	w1	w2	w3
1	0.1	0.1	0.8	19	0.3	0.4	0.3
2	0.1	0.2	0.7	20	0.3	0.5	0.2
3	0.1	0.3	0.6	21	0.3	0.6	0.1
4	0.1	0.4	0.5	22	0.4	0.1	0.5
5	0.1	0.5	0.4	23	0.4	0.2	0.4
6	0.1	0.6	0.3	24	0.4	0.3	0.3
7	0.1	0.7	0.2	25	0.4	0.4	0.2
8	0.1	0.8	0.1	26	0.4	0.5	0.1
9	0.2	0.1	0.7	27	0.5	0.1	0.4
10	0.2	0.2	0.6	28	0.5	0.2	0.3
11	0.2	0.3	0.5	29	0.5	0.3	0.2
12	0.2	0.4	0.4	30	0.5	0.4	0.1
13	0.2	0.5	0.3	31	0.6	0.1	0.3

(continued on next page)

Table B.1 (continued)

WID	w1	w2	w3	WID	w1	w2	w3
14	0.2	0.6	0.2	32	0.6	0.2	0.2
15	0.2	0.7	0.1	33	0.6	0.3	0.1
16	0.3	0.1	0.6	34	0.7	0.1	0.2
17	0.3	0.2	0.5	35	0.7	0.2	0.1
18	0.3	0.3	0.4	36	0.8	0.1	0.1

Note: w1, w2 and w3 represent the weight for rescue efficiency, flood severity and social vulnerability, respectively.

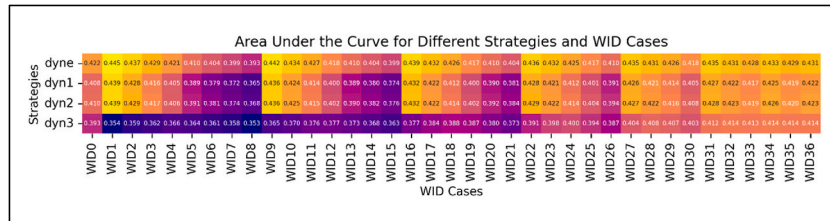


Fig. B.1. Area Under the Curve for different strategies in the case of neglecting overhead power lines, with different weight combinations. dyne, dyn1, dyn2, dyn3 represent Plan Equal, Plan 1, Plan2 and Plan 3, respectively.

APPENDIX C. Lifeboat Dynamic Allocation Illustration

To better represent the rescue operations, the allocation ratio of lifeboats among emergency facilities for each strategy dynamic varied with the rescue progress.

Figure C.1 An Illustration of dynamic allocation of lifeboat for equal plan.

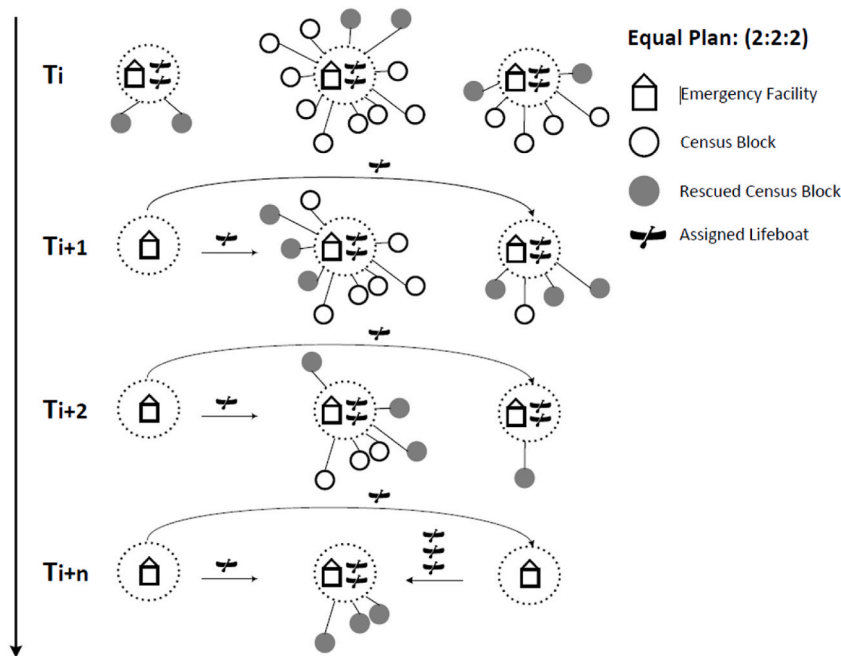


Fig. C.2. An Illustration of dynamic allocation of lifeboat for plan 1.

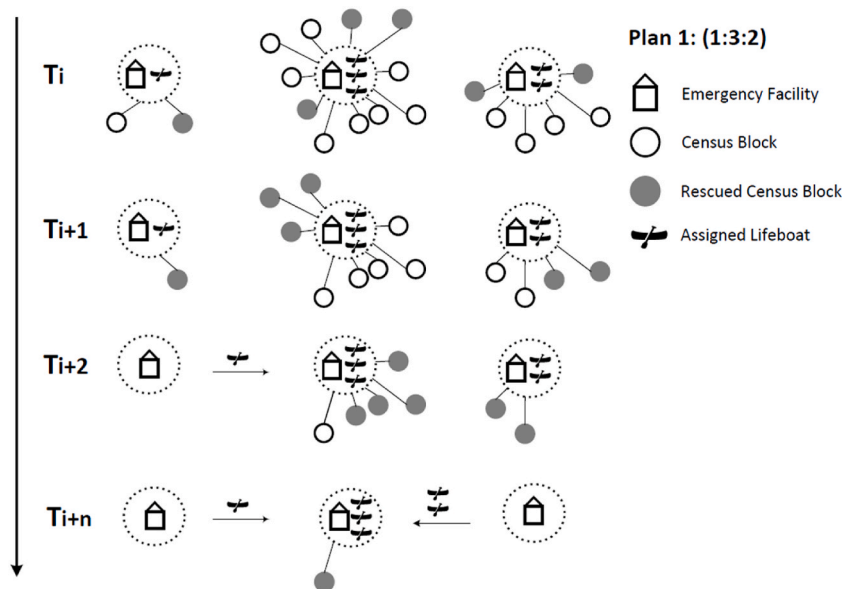


Fig. C.3. An Illustration of dynamic allocation of lifeboat for plan 2.

Data availability

Data and model will be fully open to the public. They will be uploaded to NSF Data commons.

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